

In-Class Exercises #5

The zip file of `Files4Lab5.zip` contains two java files and two images.

1. Within `Codes**8490`, create two java files in the same names as the provided files and replace `XY` with your initials.
 - The provided code follows a scene graph whose structure is the same as the one for Lab #2, which results in `CodeLab5XY.java` being almost identical to `CodeLab2XY.java`. The change of shape is introduced in `L5TextureSurfaceXY.java`.
 - Running your program should produce an unfinished table due to the missing top surface (Fig.(b)) and having only one of the four supporting legs (Fig.(a)).
 - When the menu item **Table** is selected, the (unfinished) desk top changes its shape from four to eight sides and visa versa, along with the linked single leg.
2. Modify your files to produce the result as described below.
 - (a) Complete the table's top shape by attaching the missing surface, which is mapped with the texture of an image of your choice (Fig.(c)).
hint: There is a need to first divide the original image into four quarters and then swap the diagonal components. For example, shown at the top right corner of Fig.(c) is the swapped image used to render the image shown in Fig.(c).
 - (b) Complete the table by adding three more legs in the identical shape (and appearance) as the existing leg and at locations with the same distances to their corresponding corners as well, as shown at the lower left corner of Fig.(c)
 - (c) In addition, a selection of menu item **Table** results in a change of all object shapes from 4, to 8, 12, 16 sides and back to 4 after reaching 16 (center of Fig.(c)). At the same time, the frame title updates to indicate the side number for the shape in display.
 - (d) Make your program easy to comprehend by adding adequate amount of comments.
3. Locate the folder that contains all files of your project `Comp8490`; produce a zip file of your project folder; and submit it online before due.

